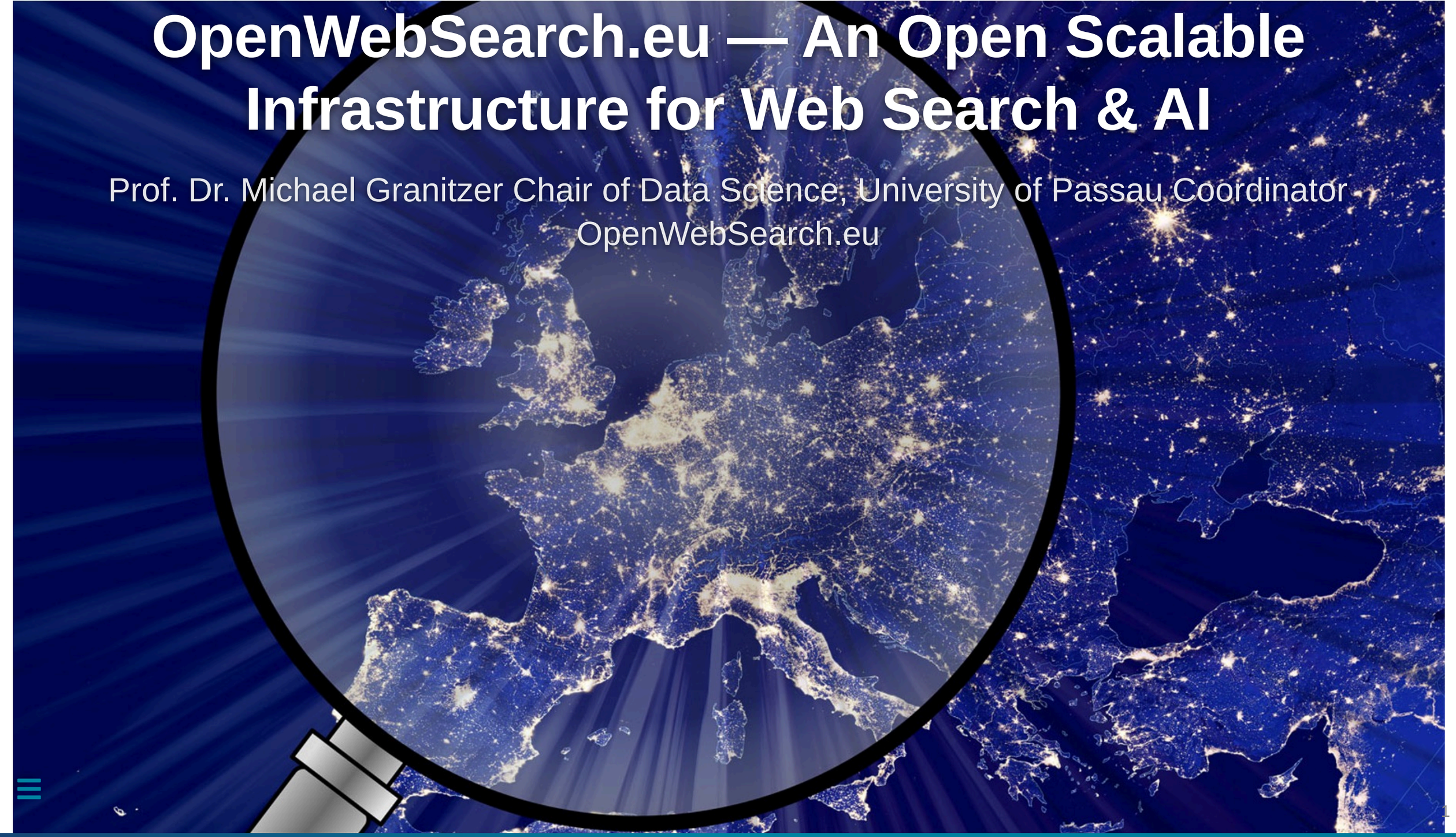
# OpenWebSearch.eu — An Open Scalable Infrastructure for Web Search & AI

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator OpenWebSearch.eu

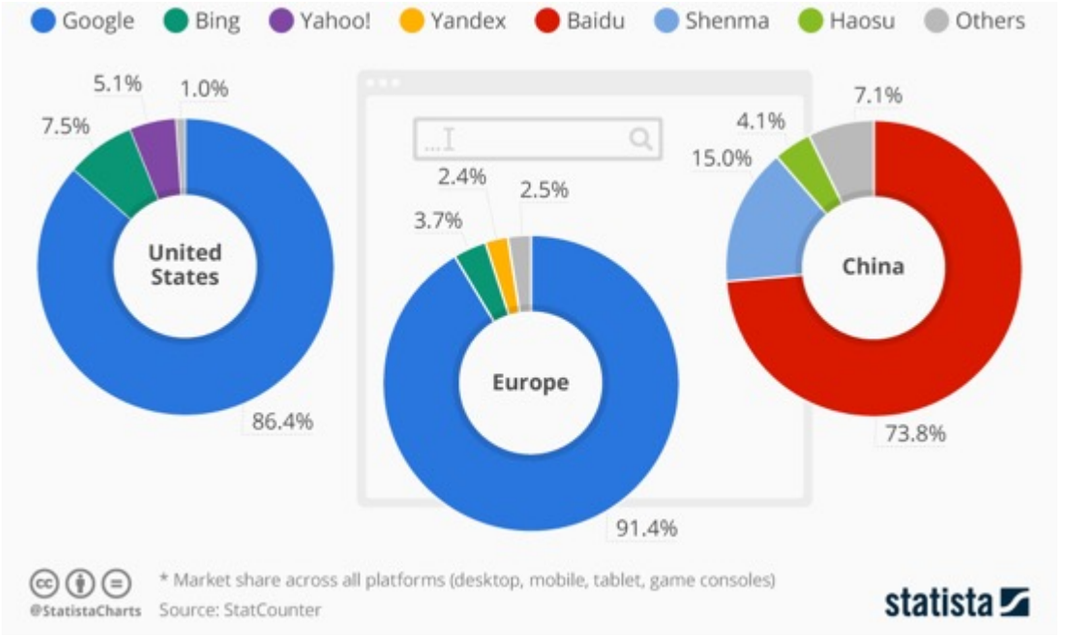


# Web Search Today

A critical resource managed as an oligopoly

# Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**
Netcraft site responses
272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**
may account for roughly half of all pages
heavy-tail intuition from large web graph studies

**~ 400 Billion**
documents in Google's search index
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**
monthly net change in Netcraft site responses
Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**
same monthly net change, averaged across December

**149 ZB (global)**
all digital data created / copied worldwide in 2024
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**
observed across websites
W3Techs usage survey

**249 CMS Platforms**
CMS platforms identified in the HTTP Archive dataset
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**
content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista

- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

# Challenges

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- High infrastructure costs
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High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



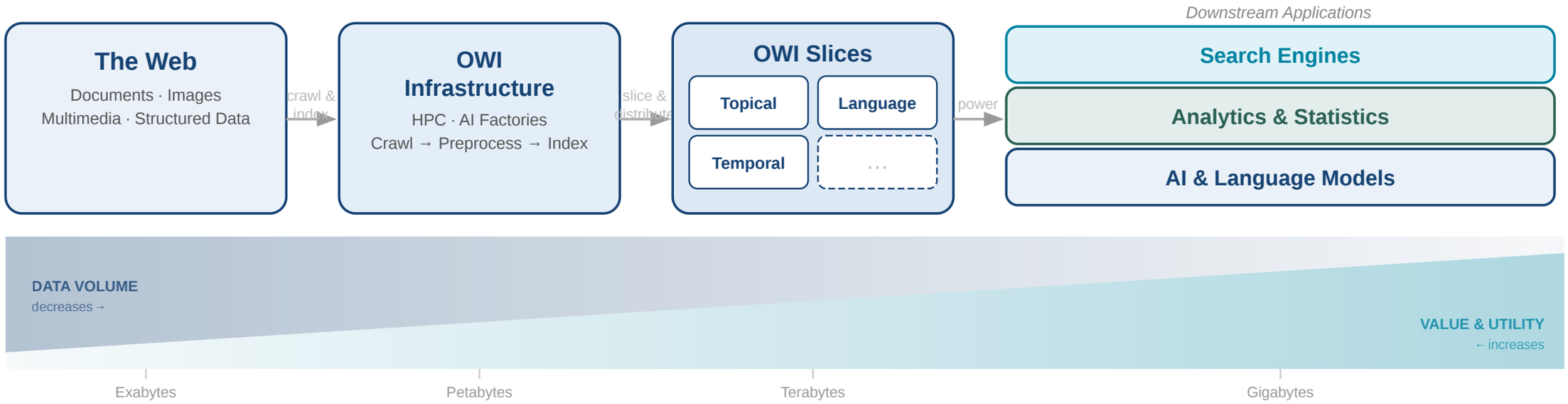

Plus 16 financially supported third-party projects (FSTPss)

Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe




*Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).*

# Conceptual Workflow

## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.

**(1) Selecting web resources**
Web pages are navigated, prioritized and collected

**(2) Storing web documents**
Multiple gatherers collect web documents and store them in web archives on a European server

**(3) Content extraction**
The content of web documents is extracted (e.g. words, images)

**(4) Metadata extraction**
Metadata (e.g. publisher, author, date) are extracted

**(5) Content analysis**
Features of web documents are extracted (e.g. topic, language, quality, genre, legal constraints, ethical aspects, etc.)

**(6) Index building**
All extracted data from web documents are stored in a specialised database, the so-called webindex

**(7) Index deployment**
The index is deployed in its full version at European data centres or sliced into smaller portions for specific purposes and made available for download

## Search Applications

A user search request will be answered by a search application that makes use of the open web index.

**(a) Selecting web documents**
Web documents are selected that fit to the user search request

**(b) Ranking web documents**
The selected Web documents are sorted (ranked) according to their assumed relevance for the user

**(c) Purpose-specific search**
An application with user interface enables the search for general or specific purposes

**(d) User searches and receives result**
The user is supported to better understand the search process

## Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications

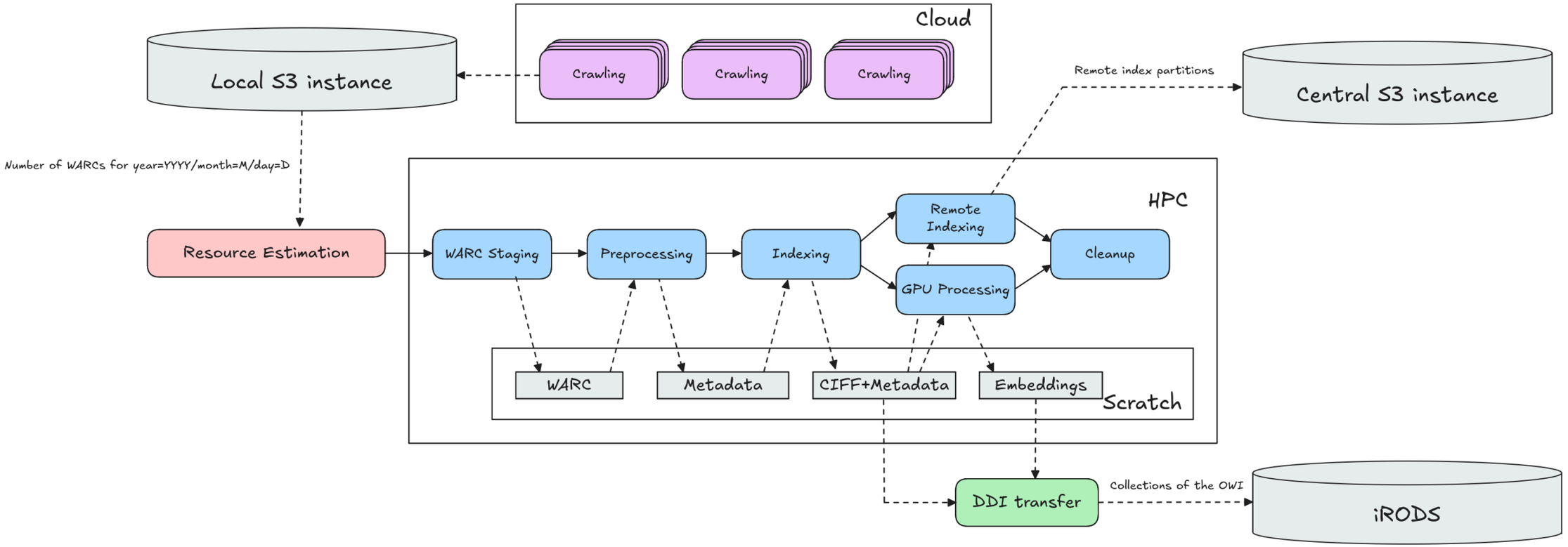
**Building knowledge graphs**
Using the extracted information from web documents, a knowledge graph is created that supports specific search requests

**Building AI Language Models**
Creation of different types of language models by using Web documents

**Any agent, multiple applications**
Language models and knowledge graphs can be used by any agent (or application)

...

# Daily Workflow Pipeline




## Crawlig Infrastructure (2026)

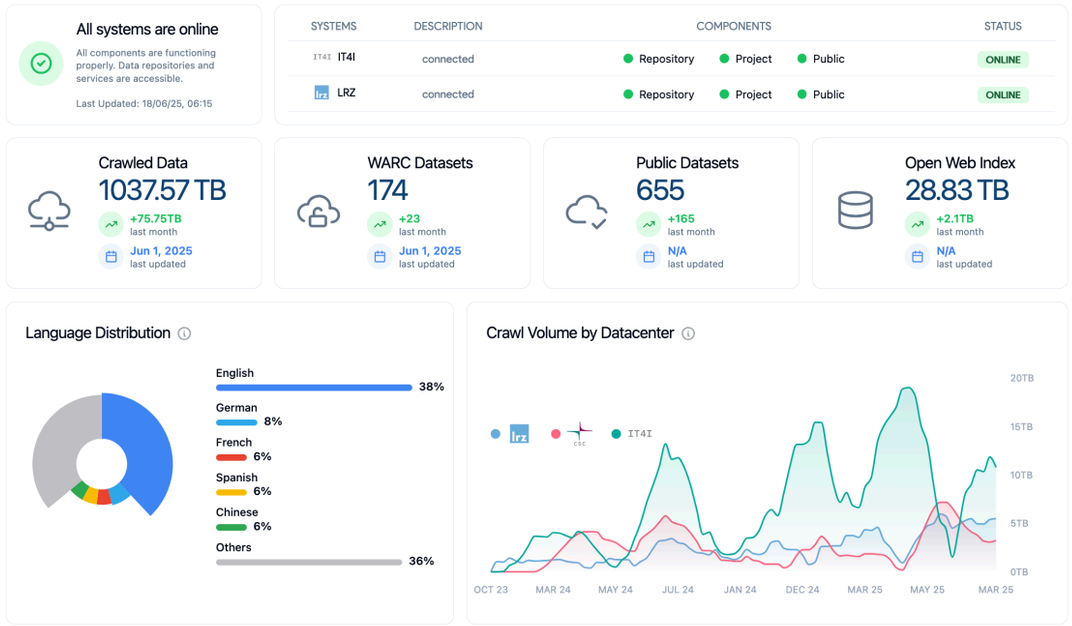
- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

# Accessing the Open Web Index

# Dashboard — Statistics & Datasets in Detail




## OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

## OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or `owilix remote pull`

# Dashboard — Corpora, Models & Search

## OWI Corpora

Curated, administrator-prepared collections for distribution:

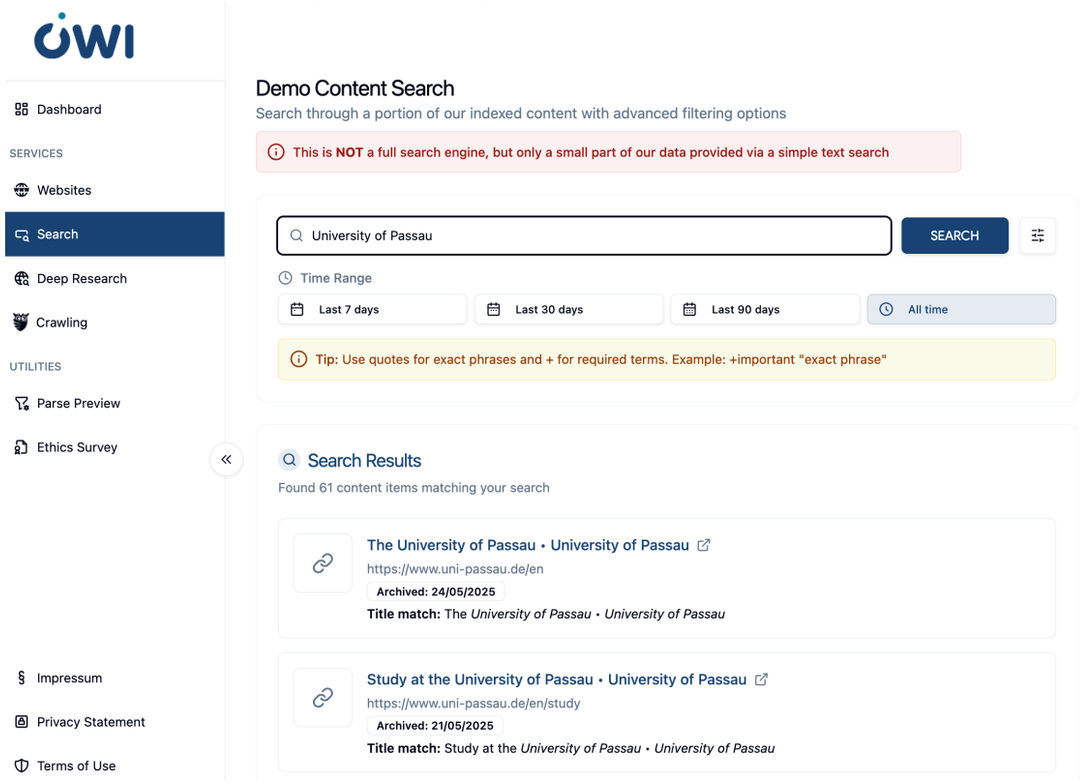
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

## OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks, and GGUF file listing

## OUR Search (Demo)

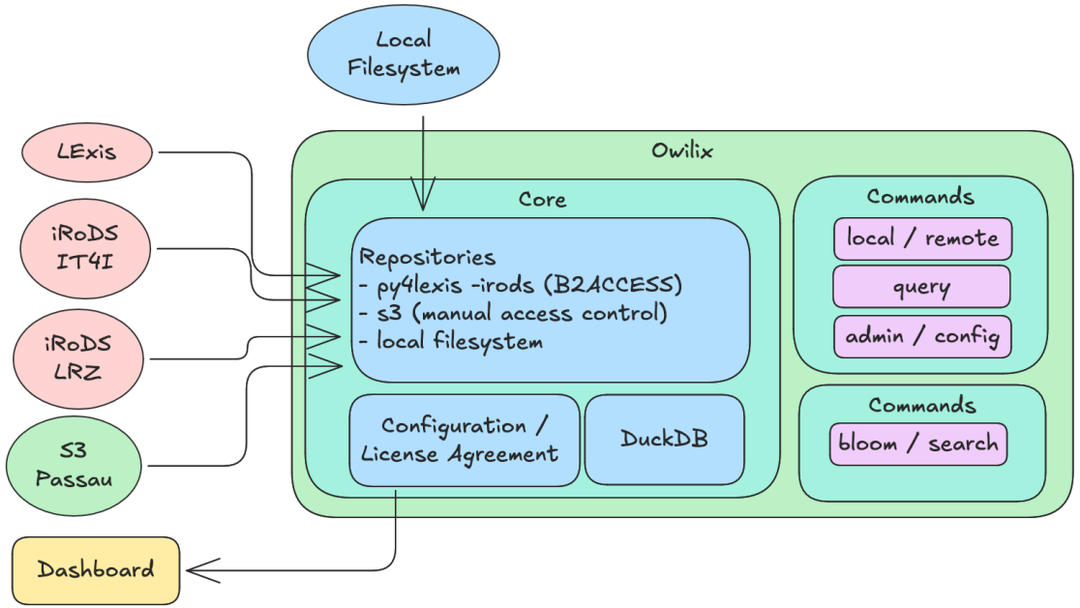



Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

## Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

# owilix — The Open Web Index Client



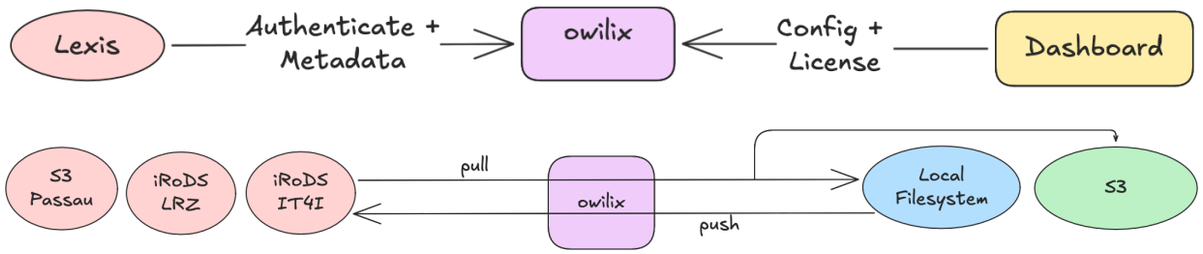

Four main workflows:

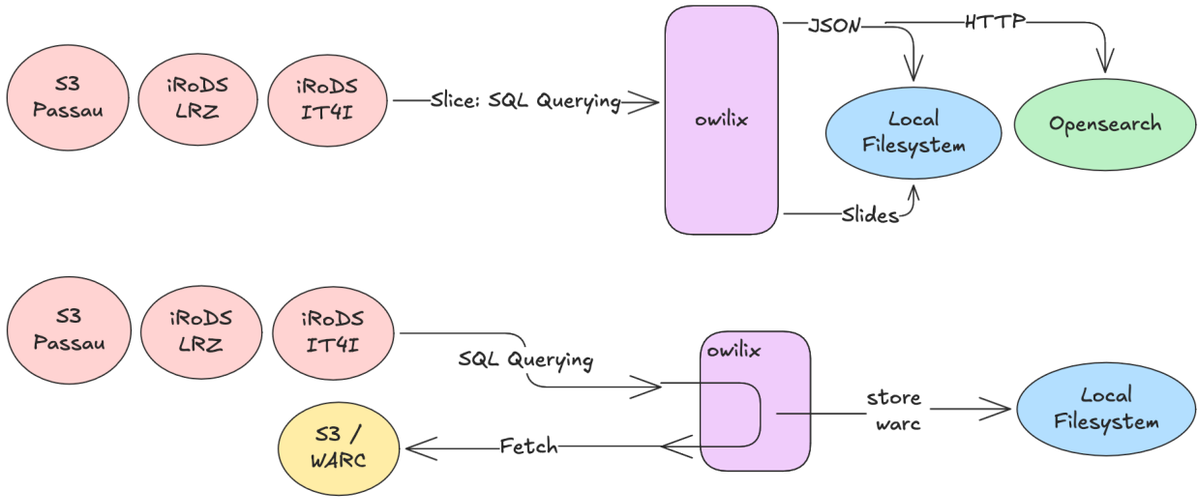
1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

Slice the index by:

- **Domain** (.de, .at, .fr, …)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

# owilix Workflows



# Remote Index — Query Without Downloading

## What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

## Example queries

```sql
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

## Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full shard download

## OUR²S — Online Search API

OUR²S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

→ OUR²S details in the dedicated OUR²S column

## Access summary

| Mode | Tool | Authentication |
|---|---|---|
| Browse | Dashboard | Anonymous / B2ACCESS |

| Mode | Tool | Authentication |
|---|---|---|
| Download | owilix | B2ACCESS (LEXIS) |
| SQL queries | DuckDB remote | B2ACCESS |
| REST search | OUR²S API | API key |
| Conversational RAG | MosaicRAG | Open / B2ACCESS |

# MosaicRAG — Building RAG Systems on OWI

### From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

### Pipeline for building your own MosaicRAG instance:

1. Download OWI slice with `owilix` (topic, domain, or language filter)
2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
3. Configure retrieval pipeline: sparse, dense, or hybrid
4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
5. Optionally add summarization and relevance marking

> Full documentation: **openwebindex.eu/book** — "Building vertical search engines with MOSAIC"

### Supported LLMs

| Task | Models |
|---|---|
| Reranking | gemma2, qwen2.5, llama3.1 |
| Summarization | + DeepSeekv3 |
| Relevance marking | gemma2, qwen2.5, llama3.1 |

### Live Demo

Try conversational search on the Open Web Index at

mosaicrag.ows.eu

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → mosaic.ows.eu
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index →

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems,

args.me/owi

- **CERN Search** — institutional search + disability knowledge search (Nooon)

and domain-specific applications

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset | Scale | Use case |
|---|---|---|
| **WebFAQ** | 96M Q&A, 75 langs | QA training, cross-lingual IR |
| **WebFAQ 2.0** | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| **Imprints** | 5.25M hosts, geocoded | Enterprise analytics, Eurostat |
| **German Commons** | 154B tokens | German LLM pre-training |
| **OWS-Curlie-2025** | 20.7M docs + qrels | IR evaluation benchmark |

All datasets available at openwebindex.eu/corpora

## Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

| | |
|---|---|
| **Website** | openwebsearch.eu |
| **Dashboard** | openwebindex.eu |
| **Code Repository** | https://opencode.it4i.eu/openwebsearcheu-public/ |
| **Community** | openwebsearch.eu/community |
| **Join** | join@openwebsearch.org |
| **General** | community@openwebsearch.eu |

**Thank you — Questions, Comments, Feedback very welcome!**